



Study of an Adaptive Financial Recommendation Algorithm Using Big Data Analysis and User Interest Pattern with Fuzzy K-Means Algorithm

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Abstract

In an ever-changing financial market, big data is set to revolutionize user interest management by sparking innovation and reshaping recommendations for the future. Conventional financial services face significant challenges like accessibility, personalization, limited reachability, and incomplete information about user interest patterns. Thus, it results in suboptimal financial recommendations that do not fully capture the individual user interests and adapt to changing market conditions. Hence, the research developed an adaptive algorithm that uses fuzzy logic, neural networks, and big data to deliver accurate financial recommendations based on user patterns using the Fuzzy Neural Financial Recommendation (FNFinRec) Algorithm. Implemented on a Hadoop platform with a MapReduce framework, the FNFinRec ensures efficient processing of large datasets. Fuzzy K-means clustering is applied, in which fuzzy logic helps to handle uncertainties in financial data and clusters users with similar patterns. An adaptive user profile is developed based on real-time user data input. The Neutral Collaborative Filtering (NCF) recommendation approach aims to predict user interests in financial products/services by learning from user interaction data. The neural networks provide personalized financial recommendations, adapting to changes in user patterns over time for improved accuracy. The metrics such as silhouette coefficient, Davies–Bouldin Index, mean square error (MSE), Precision@k, and Recall@k are used to assess the algorithm's performance with existing algorithms. The results show that the proposed FNFinRec algorithm outperforms existing methods regarding clustering quality and recommendation accuracy. The competitive processing times that FNFinRec achieves are also crucial for making real-time financial decisions.

Keywords Financial recommendation · Big data analysis · Fuzzy K-means algorithm · User interest pattern

1 Introduction

The rise of big data has brought forward an evolutionary change in the constantly evolving financial sector, requiring a comprehensive understanding of market conditions and user recommendations [1]. The user experience and operational effectiveness in financial services will be significant. Incorporating big data to handle large-volume transaction platforms with existing systems can create new business opportunities [2]. One of the most successful recommendation systems uses collaborative filtering (CF), which predicts user interest by collecting data from former users with

identical perceptions [3]. The adoption of AI in extensive data analysis emphasized the need for robust monitoring and adaptation to the evolving changes in the financial sector and technological landscape [4]. Machine learning and vast amounts of data have accelerated the development of investment recommendation systems and demonstrated how these innovations can improve users' stock fund management and financial literacy [5].

Big data analysis enables financial service providers to provide personalized services by analyzing user interests and spending patterns. However, several challenges must be focused on to fully adapt the use of big data analysis and ML in financial services [6]. Financial sectors share massive amounts of data daily, with big data analysis impacting the productivity of services [7]. User-friendly experience-based financial services adopt AI techniques and big data analysis to reshape user disparities and financial growth opportunities [8]. Earlier K-means clustering algorithm poses challenges

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with large-scale data and noise due to its preset user-related values and sensitivity to initial centroids. It also discussed improvement using mean shift technique to focus on big data analysis [9]. With this clustering issue, the user groups are clustered using adaptive learning-based improved k-means clustering with an additional optimization algorithm for cluster centres [10]. The study demonstrates its progressive nature in the test of real financial data, reaching a clustering accuracy of over 70% significant recommendation for subsequent financial analysis and user data management [11]. Finance recommendation system identifies user interests' patterns and supplements for decision making of a financial stock trader but still needs improvement towards accurateness of user pattern analysis [12]. However, the recommendation algorithm that solely considers recommendation diversity and accuracy is insufficient given the significance of the growing amount of user activity data and individual user interests [13].

With the use of real-time relevant feedback and user activity data, the recommendation systems may automatically create profiles that can identify users' actual interests and make recommendations about their preferred patterns [14]. The study applies fuzzy k-means clustering for segmenting the users based on their interests in frequency, time and location adaptability for achieving financial goals [15]. For obtaining more accurate results, user pattern in clustering formation is analyzed with the optimal cluster numbers and initial values of its centers [16]. The study proposes an automatic recommender system for investment types using adaptive fuzzy k means algorithm with neural architecture based on user feedback and improves decisions [17].

Novel contributions:

- (1) To process and analyze large volumes of financial data and facilitate distinct user investment patterns, employs MapReduce framework on Hadoop platform.
- (2) To handle uncertainty in user preferences by clustering users based on their interest pattern using fuzzy k-means algorithm.
- (3) To analyze the user interest patterns thereby providing precise predictions and adaptive financial recommendations using neural collaborative filtering (NCF) approach.

Manuscript order: A summary of reviewed literature work of existing recommendation techniques related to financial data based on user interest is discussed in Sect. 2. Section 3 deliberates the data collection from financial sources. Section 4 discusses Fuzzy Neural Financial Recommendation (FNFinRec) Algorithm for adapting to the changing user interest pattern. Section 5 delves deeper into the implementation of comparison analysis of clustering and big data analysis, while Sect. 6 concludes the research work.

2 Overview of Existing Studies

2.1 Traditional Recommendation Algorithms

Luo [18] presented an optimized incremental clustering approach assisted using kernels for providing investment recommendation for financial users. The experimental dataset for the study is based on stock from the Shanghai Stock Exchange. The results show that the suggested algorithm helps stabilize markets and decreases risks associated with financial investments. However, the possibility of scaling issues with bigger datasets and the requirement for additional processing to account for ever-changing market conditions are needed.

Asem et al. [19] applied users' feedback, fuzzy logic, and neural network inference to improve investment recommendations in the adaptive neural fuzzy inference system (ANFIS) model. The study makes use of demographic profile and financial types to give adaptive recommendations by applying data-driven algorithms. A web-based collection of potential investors' demographic and investment data is used by the program to process seven main parameters. The JMP and MATLAB system groups investors and makes customized investment suggestions. The model's dependence on precise initial gathering of data and potential scaling concerns with larger, more diversified datasets are drawbacks.

Sengupta et al. [20] presented a portfolio recommender system (PRS) for financial service optimization using Markowitz's Modern Portfolio Theory (MPT), a greedy algorithm, and integer programming. For FY 2021–22, the dataset includes NSE closing prices for the day and transaction data for 297 Indian equities. K-means clustering finds dissimilar equities and optimizes portfolios, with greedy algorithm surpassing integer programming in financial resource management. The model relies on past data, which may not accurately predict market trends or investor behavior.

2.2 Big Data Analysis and Its Applications in Finance Based on User Interest Patterns

Li et al. [21] suggested smart investing platform evaluation, with a particular emphasis on China Merchants Bank's Financial Capricorn Intelligence, for their investment efficiency using the K-means clustering algorithm. This platform's investment preferences of users data forms the basis of the dataset. According to the findings, Capricorn Intelligence's investing approach is timing-oriented, and its business shows signs of unpredictability in choosing funds and a tendency to lower returns to

control risk. The study constraints include limited data mining depth and possible algorithmic model discrepancies that could lead to an inaccurate assessment of investment effectiveness.

Dandugala and Vani [22] The Improved Fuzzy C-Means Multiview technique (IFCM) clustering technique uses the MobileNet V2 model and a Three-Layered Stacked Bidirectional LSTM (3L-BiLSTM) to speed up huge market data analysis. MapReduce is used to implement the method on Hadoop to improve efficiency and reduce time complexities. Research on three typical data sets showed greater performance in different parameters compared to previous techniques. The study may be constrained by the computing capabilities needed for large-scale processing of data and the assessment datasets, which may not apply to other big data.

You [23] introduced a state-of-the-art models in marketing changes with E-commerce recommendation method that surpasses the shortcomings of earlier systems by combining large data analysis with fuzzy clustering gene analysis. Integrating genetic fuzzy analysis to improve recommendation accuracy is the main contribution. As a result of its well-configured algorithm, the mean absolute error (MAE) is minimal. The complexity of dealing with large-scale data and the necessity for additional optimization in real-time processing, however, limit its performance.

Chiou-Wei and Lee [24] applied a financial recommendation model called K-medoids clustering using Kullback–Leibler-K-Medoids (K-L-KM) divergence for fund recommendations, algorithm. User-item rating matrix was employed to identify same funds based on their similarity measured using an enhanced KL distance cluster. KL similarity is 0.85 accurate on average and RMSE ranges from 0.93 to 0.94. Data complexity can be reduced while important patterns are preserved using principal component analysis (PCA). Possible sensitivity to clustering variables and limitations on dataset size are its limitations.

Concerns with data quality, scalability, and flexibility in real time are among the obstacles encountered by current research on financial recommendation systems. It might be difficult for traditional methods to deal with massive datasets and adjust rapidly to changes in the market. To tackle these issues, the FNFinRec technique uses a MapReduce framework on Hadoop for scalable processing, fuzzy K-means clustering for data uncertainty management, and neural collaborative filtering for real-time accuracy and adaptability. By boosting suggestion accuracy and system efficiency, this integrated strategy improves efficiency in dynamic financial environments, overcoming limits of conventional methods.

3 Research Analysis

Research in financial management and clustering has recently made great strides, with an emphasis on improving clustering performance and quantifying financial data indicators. The current research aims to provide intelligent financial recommendation model based on user interest pattern. Fuzzy K-means clustering can accurately aggregate vast volumes of user profile into clusters depending on financial investment preferences and behaviors depicted in Fig. (a and b). NCF may then exploit these clusters to deliver adaptive recommendations, despite the setting of massive financial data sets. The utilization of algorithms that use deep learning such as NCF and the integration of different features show that the data can manage the complicated and varied nature of big data analysis. A distinct form of data sources from various big data analysis is validated using silhouette coefficient.

Table 1 indicates key user attributes for the financial recommendation system are summarized from the source [25]. Gender breakdown, age range, investment characteristics, and projected returns are all part of it. The table gives a thorough picture of user profiles and habits by including graphical representations of investment length and monitoring frequency.

3.1 Data Preprocessing and Feature Selection

The initial data inspection selects the relevant features that influences the financial decisions and user interest patterns. The set of users can be represented in the form of mathematical evaluation given in Eq. (1).

$$U = \{u_{r1}, u_{r2}, \dots, u_{rm}\} \quad (1)$$

Followed by defining the set of items called financial products/services offered to the users can be defined as:

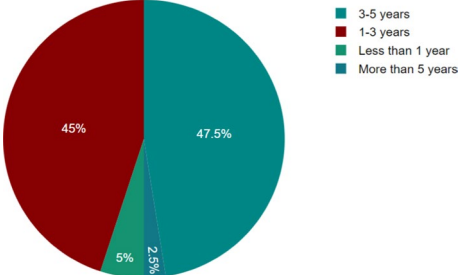
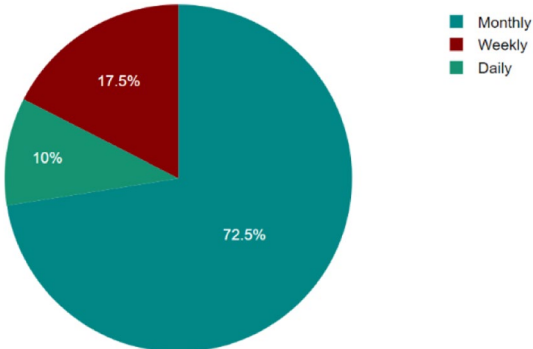
$$I = \{i_1, i_2, \dots, i_n\} \quad (2)$$

The interaction matrix R is defined using Eq. (3).

$$R_{m \times n} = \begin{bmatrix} r_{11} & \cdots & r_{1n} \\ \vdots & \ddots & \vdots \\ r_{m1} & \cdots & r_{mn} \end{bmatrix} \quad (3)$$

$r_{u,i}$ indicates the interaction between user u_r and item i . For analyzing the user interest pattern, the financial risk score Ψ is calculated based on the investment preferences P_i and expected returns E_i , a simple weighted sum can be used. The other attribute investment diversity namely an investment diversity score D can be calculated using the entropy

Table 1 Financial profile overview

Attributes	Distributions
Gender distribution	Female 43.1%, Male 56.1%
Age category	21–35
Features used	Mutual funds, equity market, debentures, government bonds, fixed deposits, PPF, gold
Investment duration	Duration  Fig (a) Investment duration analysis
Investment monitoring frequency	Invest_Monitor  Fig (b) Investment monitoring frequency analysis
Expected return	10–20%, 20–30%, 30–40%

of the distribution of preferences across different investment avenues analyzed in Eq. (4).

$$D = - \sum_{i=1}^n p_i \log(p_i) \quad (4)$$

where p_i is the proportion of investment in the i^{th} option that may include stocks, bonds, real estate, mutual funds, and gold. A more diverse investment, with a greater entropy value, is crucial for adaptive financial recommendations that aim to balance risk and return based on user expectations and observed patterns. Feature selection involves the process of identifying the importance of evaluated features using random forest concept where the importance of feature j in a random forest can be calculated using Eq. (5)

$$I_j = \frac{1}{T} \sum_{t=1}^T I_j^{(t)} \quad (5)$$

where $I_j^{(t)}$ is the importance of feature j in tree t , and T is the total number of trees in the forest. This step helps in identify the most influential factors in predicting user investment preferences. The suggested FNFinRec algorithm's whole process flow is illustrated in this Fig. 1. Starting with the intake of data and progressing via preprocessing, grouping, and NCF, it culminates in the output of recommendations.

3.2 Big Data Analysis

Processing and analyzing massive amounts of financial data are much improved when the MapReduce framework is used on the Hadoop platform. Distributing processing duties across numerous machines provides scalable data handling with MapReduce. The analysis is essential for managing massive datasets. The framework's "Map" and "Reduce" stages divide data processing, enabling parallel computation and result aggregation to speed up the recommendation system and clustering model training in a qualified manner.

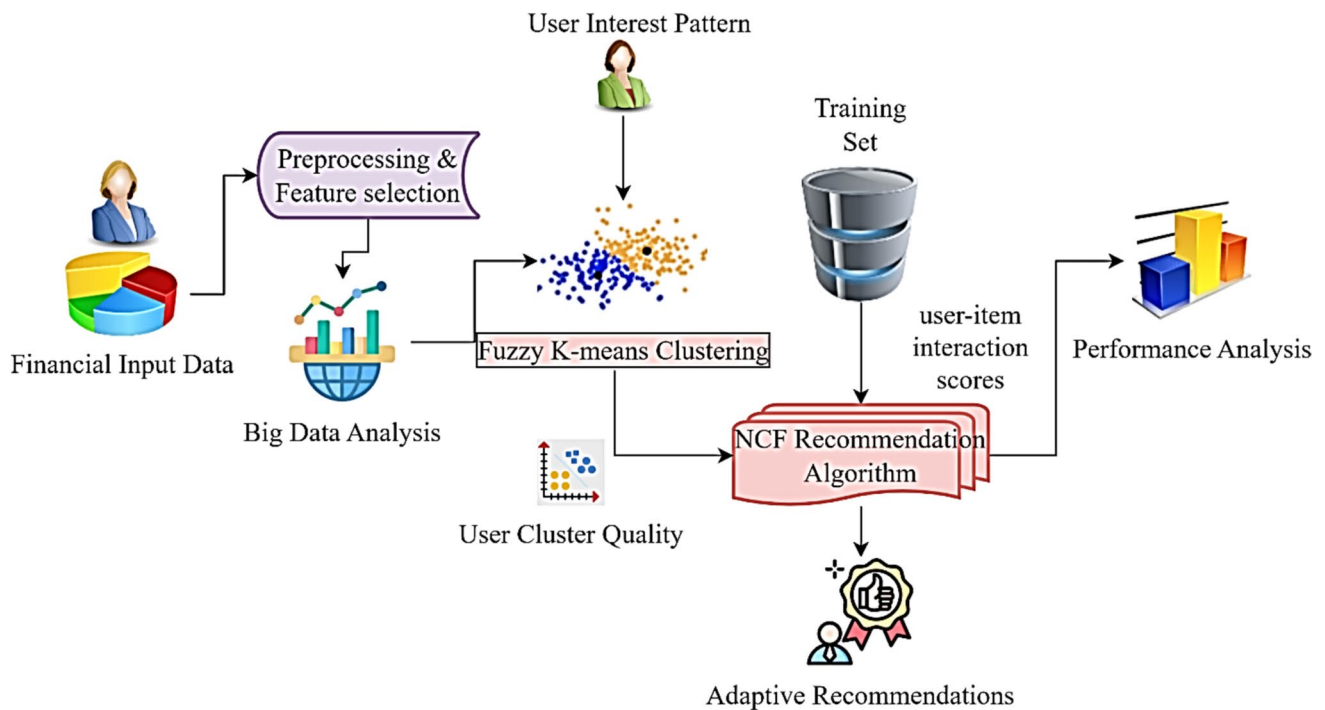


Fig. 1 Overall working flow of fuzzy neural financial recommendation (FNFinRec) algorithm

Hadoop guarantees quick data processing by effectively managing resources and organizing activities. The research analysis is crucial for real-time updates and synthesizing feedback from users. Because it is more adept at dealing with complicated data, this method makes data preprocessing more efficient and makes financial recommendations more accurate.

3.3 Fuzzy K-Means Clustering

Fuzzy K-Means is helpful as it organizes financial resources according to factors like risk levels and expected returns, allowing it to identify patterns in the relationship between the two. The users with conservative tolerance for risk can be suggested low-risk, stable-return investments that work based on personalized financial plans that match user preferences. Applying fuzzy logic to handle uncertainty in user preferences and data, providing adaptable recommendations by clustering users with similar financial patterns. The fuzzy k-means clustering minimizes Within-cluster variance with the following objective function

$$J_m = \sum_{k=1}^K \sum_{i=1}^m u_{rik}^m \cdot \|x_k - c_k\|^2 \quad (6)$$

In Eq. (6), the variable m indicates the fuzziness parameter typically $m > 1$, u_{rik} is the degree of membership of

the datapoint x_i to cluster k and c_k indicates the centroid of cluster k

Step 1: define appropriate membership functions for fuzzy sets based on user attributes.

Due to the subjective character of users' u_r financial decisions, membership values will show the extent to which each user is an integral component of specific clusters. The degree to which clusters overlap is controlled by the fuzziness parameter m . There is more overlap when m is larger, which means that users can be members of more than one cluster. Each data point x_i represents an individual user's financial profile. This profile could include features like age, gender, preferences for various investment avenues that involve mutual funds, equity market, debentures, and other factors such as investment objectives, expected returns, and investment duration. The membership value u_{ij} indicates how much the i^{th} data point called the user profile belongs to the j^{th} cluster it is calculated using an Eq. (7).

$$u_{rij} = \frac{1}{\sum_{k=1}^K \left(\frac{\|x_i - c_j\|}{\|x_i - c_k\|} \right)^{\frac{2}{m-1}}} \quad (7)$$

The parameter $\|x_i - c_j\|$ indicates the Euclidean distance between the data point x_i and the centroid c_j of cluster j with considered features like investment preferences. From this, the hypothesis may provide if user i heavily invests in mutual funds and centroid j represents a high-risk investment cluster,

the distance might be large, reflecting that the user is far from the profile typical of that cluster. The ratio $\left(\frac{\|x_i - c_j\|}{\|x_i - c_k\|}\right)$ compares the distance of the datapoint x_i to the centroid of cluster j with its distance to the centroid of another cluster k . The considered hypothesis is if user i is closer to the centroid of a low-risk cluster c_j compared to a high-risk cluster c_k , this ratio will be smaller, indicating a higher membership value for the low-risk cluster. The exponent parameter $\frac{2}{m-1}$ where $m > 1$, which controls the degree of fuzziness, is used to adjust the influence of the distance ratio based on the m . The method takes

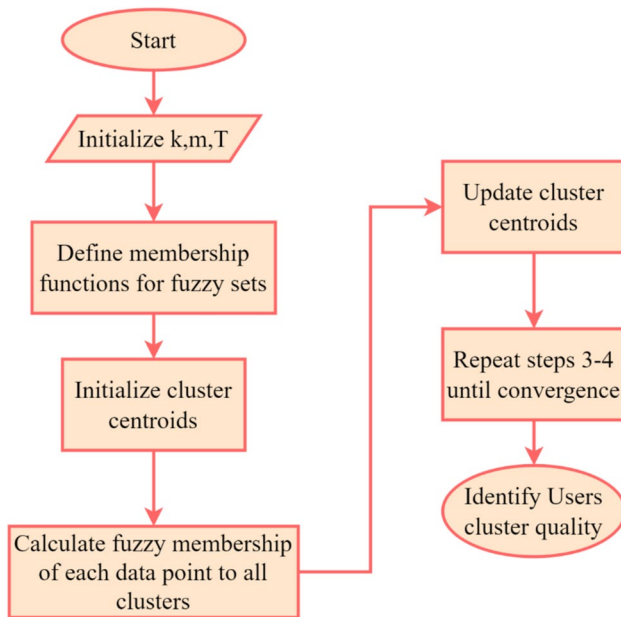


Fig. 2 Flowchart of fuzzy k-means clustering algorithm

uncertainty with $m > 1$ into consideration and permits users to have varied degrees of membership in multiple clusters, mirroring the complexity of financial actions in the real world. The sum in the denominator ensures that the membership values across all clusters for a given data point sum to 1, reflecting the probability-like interpretation of these values.

Step 2: initialize cluster centroids randomly.

Initialization of centroids starts with the random or informed representing investment profiles with a different focus on users preferring mutual funds and other favorites on the equity market. Adjusting the clustering to represent user financial preferences for various financial services is the goal of the clustering criterion. For instance, if a user has a strong preference for equities investments, their interests may be grouped. However, some overlap may exist if they demonstrate interest in other investment types. More effective convergence and meaningful clusters can result from initial centroids influenced by known user preferences.

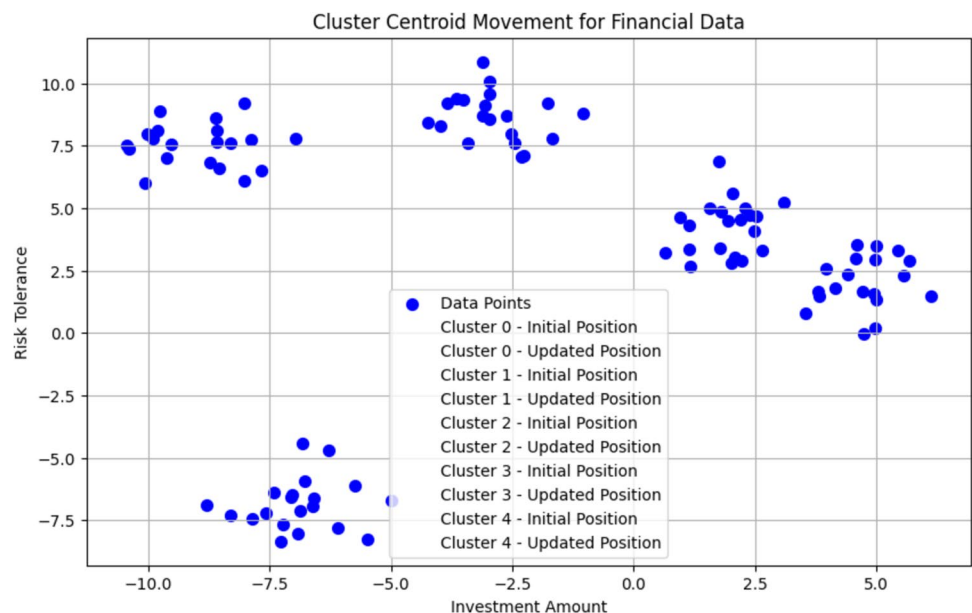
Step 3: calculate the fuzzy membership of each data point to all clusters, and the Fuzzy K-means clustering technique is broken down into its parts in this flowchart in Fig. 2.

Step 4: update cluster centroids based on fuzzy memberships.

The centroid c_j of cluster, j is updated based on each user's investment pattern with a degree of membership to different clusters. The membership value u_{rij} indicates how strongly a user i belongs to cluster j using this Eq. (8) to reflect their investment behavior.

$$c_j = \frac{\sum_{i=1}^n u_{rij}^m x_i}{\sum_{i=1}^n u_{rij}^m} \quad (8)$$

Fig. 3 Cluster centroid update



where u_{rij} in Eq. (8), is the membership value of the data point x_i to cluster j , m as a fuzzy parameter determines how clusters overlap and how membership values are calculated and n as number of data points is shown in Fig. 3.

Step 5: repeat steps 3 and 4 until convergence that indicates a change in centroids between iterations is minimal and below a threshold.

$$\Delta c_j = \|c_j^{(new)} - c_j^{(old)}\| \quad (9)$$

where Δc_j in Eq. (9) should be below the predefined threshold to indicate convergence.

Step 6: analyze resulting clusters to identify distinct user groups based on financial interest. The time complexity of fuzzy k-means is generally $O(T \times K \times n \times d)$ where T indicates the number of iterations, K represents the number of clusters, n as the number of data points, and d is the dimensionality of the data. The cluster quality is analyzed using silhouette score and David Bouldin Index (DBI) to measure the average similarity ratio of each cluster with its most similar cluster. The FNFinRec algorithm generates synthetic data, applies fuzzy K-means clustering, and calculates the silhouette score to evaluate clustering outcomes, essential for segmenting customers by financial preferences.

$$DBI = \frac{1}{k} \sum_{i=1}^K \underbrace{\max}_{j \neq i} \left(\frac{S_i + S_j}{d(\text{cent}_i + \text{cent}_j)} \right) \quad (10)$$

$$S_i = \frac{1}{|C_i|} \sum_{x \in C_i} d(x, \text{cent}_i) \quad (11)$$

where K in Eq. (10) indicates the number of clusters and computes the ratio of the sum of scatter within clusters i and j to the distance between their centroids for each pair of clusters with the index as the average of maximum ratios. The term C_i indicates the number of data points in cluster i , x represents a data point and $d(x, \text{cent}_i)$ in Eq. (11) is the distance between x and the cluster centre C_i . For each pair of clusters, M_{ij} with i and j , calculate the distance between the centroids $d(\text{cent}_i + \text{cent}_j)$ in Eq. (12).

$$M_{ij} = \|\text{cent}_i - \text{cent}_j\| \quad (12)$$

For various cluster numbers with a minimum of two clusters, this FNFinRec algorithm computes the *DBI*, which optimizes financial data clustering. Analyzing the $S_i + S_j$ data helps segment and assist with risk management, personalized recommendations for financial services, and adaptability of financial products and services to different types of users. By incorporating the proposed clustering algorithm and Davies–Bouldin Index analysis into the big analysis platform with financial dataset, the algorithm can classify investors into different categories, enabling it to provide highly personalized financial recommendations based on $\frac{1}{|C_i|} \sum_{x \in C_i} d(x, \text{cent}_i)$. As the number of clusters increases, the trend line decreases in the Davies–Bouldin Index (DBI), suggesting that the clustering using the Fuzzy k means algorithm is becoming more successful. Improving the model's capacity to group similar financial data points, a lower DBI signifies better-defined cluster groups based on user interest patterns. This means that as the number of clusters grows, the clusters get more distinctive to provide investment recommendations based on the dynamically changing behavior of user interests.

3.4 Neural Collaborative Filtering for Adaptive Financial Recommendations

The recommendations will remain relevant since the model can adjust to user interests that change over time through continuous learning from new data. To provide users with adaptive recommendations for investments that are appropriate for their risk-return profile, the NCF algorithm learns through user data.

Step 1: user-item interaction matrix from transaction history.

Construct a matrix where rows represent users and columns represent investment avenues like mutual funds, equity markets, etc. Entries are binary, indicating whether a user is interested in an investment type (1) or (0) otherwise. This matrix is given as input to the NCF model, and the data collected from interactions enables the model to understand user interest patterns and estimate their subsequent investment patterns.

Table 2 User-item interaction matrix

User	Fixed deposits	Mutual funds	Equity	Bonds	Real estate	Crypto
1	1	0	1	0	0	0
2	1	1	0	1	0	0
3	0	1	1	0	1	0
4	1	1	0	0	0	1
5	0	0	1	1	0	0

For each user, mapping user interest patterns as given in Table 2 helps to analyze the risk levels of the investment avenues that interacted with 1 in the interaction matrix and are mapped to the appropriate risk category. Suppose a user has interacted with multiple investment avenues within the same risk level. In that case, one is selected based on predefined rules, such as the frequency of interaction and preference for specific avenues. If a user has expressed interest in certain investments, the interaction matrix will represent that interest as binary data. The level of risk associated with each investment is classified as low for fixed deposits and bonds, medium for real estate and mutual funds and high for equity and cryptocurrency. To maintain fair representation even in cases where some data are absent, the generated table shows which investment paths each user prefers within each risk category. The proposed NCF approach relies on this classification to provide adaptive financial recommendations based on users' risk-return decisions.

Users' chosen investment alternatives are displayed in Table 3 according to their low, medium, or high-risk levels. NCF analysis of user interaction data is the basis for the preferences. Each user's investing options are classified accordingly based on their comfort level with risk. Investors' risk tolerance is defined in this table as either low, medium, or high. It shows how the algorithm sorts investment opportunities according to risk profiles and how user preferences are mapped to these groups. This categorization helps make suggestions to consumers that are suitable for their level of risk.

Step 2: general matrix factorization and multi-layer perceptron for training.

They are implementing a feedback mechanism to collect user responses to recommendations, periodically update the clusters based on new user data, and adjust recommendations based on real-time market conditions and user feedback. The NCF model typically consists of an embedding layer followed by multiple dense layers where the embedding layer maps each user and item to a lower-dimensional space.

i) User embedding matrix

These embeddings aim to map users and items to a low-dimensional space where similar users and items are closer

together. This user embedding matrix can be represented in the form of $P \in \mathbb{R}^{m \times d}$ where each row p_{u_r} is a d-dimensional embedding for the user u_r .

ii) Item embedding matrix

The matrix represents each financial service or investment item in the same latent space. Similarly, $Q \in \mathbb{R}^{n \times d}$ indicates a financial service embedding matrix where each row q_i is a d-dimensional embedding for item i . The embedding size controls the dimensionality of the space, affecting the model's ability to capture complex patterns in user behavior.

Interaction prediction: The interaction prediction for the user u_r and financial item i is given by embeddings from the matrices P and Q . The interaction prediction $\hat{r}_{u,i}$ is given in Eq. (13),

$$\hat{r}_{u,i} = f(p_{u_r}, q_i) \quad (13)$$

where $f(p_{u_r}, q_i)$ in Eq. (14) can be modeled in various ways. For instance, using a multi-layer perceptron (MLP)

$$f(p_{u_r}, q_i) = \text{MLP}(f(p_{u_r} \oplus q_i)) \quad (14)$$

Here, an operator $\oplus$ indicates the concatenation, and MLP is a fully connected neural network that takes the concatenated user and financial item embeddings as input and outputs and the prediction interaction score $\hat{r}_{u,i}$.

Training with loss function: The model is trained using a loss function such as binary cross entropy for implicit feedback or MSE for explicit feedback. This loss function $\mathcal{L}(Y_{u,i})$ is used to train the model to predict user interest patterns based on implicit feedback. The goal is to minimize the loss so that the model's predictions $\hat{r}_{u,i}$ closely match the actual user interest $r_{u,i}$. For implicit feedback the binary values are considered indicating whether a user has shown interest or invested in certain investment avenues can be integrated as given in Eq. (15).

$$Y_{u,i} = \begin{cases} 1 & \text{if user } u_r \text{ has invested in item } i \\ 0 & \text{otherwise} \end{cases} \quad (15)$$

$$\mathcal{L}(Y_{u,i}) = \sum_{(u_r, i) \in R} [r_{u,i} \log(\hat{r}_{u,i}) + (1 - r_{u,i}) \log(1 - \hat{r}_{u,i})] \quad (16)$$

The representation of R indicates the set of all user and financial service item pairs provided with feedback. The term $r_{u,i}$ indicates the actual feedback given by the u_r for financial service item i followed by $\hat{r}_{u,i}$ predicted feedback for u_r for financial service item i . The term $\log(\hat{r}_{u,i})$ and $\log(1 - \hat{r}_{u,i})$ in Eq. (16) represent the logarithmic loss for correct and incorrect predictions; this loss function penalizes the model more when the projections are confident but incorrect. The hidden layers learn the interactions between user and item embeddings. Additionally, the output layer

Table 3 User investment preferences analyzed by risk level

User	Investment avenue (Low)	Investment avenue (Medium)	Investment avenue (High)
1	Fixed deposits	Mutual funds	Equity
2	Bonds	Mutual funds	Equity
3	Fixed deposits	Bonds	Mutual funds
4	Fixed deposits	Mutual funds	Crypto
5	Bonds	Fixed deposits	Mutual funds

produces a score representing the predicted interest in an investment type. The explicit feedback reflects user interest level, which can be observed from Eq. (17).

$$\mathcal{L} = \frac{1}{|R|} \sum_{(u_r, i) \in R} (r_{u_r, i} - \hat{r}_{u_r, i})^2 \quad (17)$$

3.5 Adaptive Recommendation System

The adaptive financial recommendation system guides users to make informed investment decisions using data-driven insights from fuzzy K-means clustering and NCF. The measures of the system's performance, such as recall and precision at the top N recommendation, show how well the recommendations are employed for each user, contributing credence to data-driven decision-making.

Top-N recommendation: For each user u_r , generate the top-N recommendations by sorting items based on the predicted ratings $\hat{r}_{u_r}$. Precision and recall for a user u_r at rank, N can be calculated using Eq. (18).

$$Top - N(u_r) = \text{argsort}_i(\hat{r}_{u_r})[: N] \quad (18)$$

$$pre@N = \frac{|\{rec_{items}\} \cap \{rel_{items}\}|}{N} \quad (19)$$

$$Recall@N = \frac{|\{rec_{items}\} \cap \{rel_{items}\}|}{|\{rel_{items}\}|} \quad (20)$$

4 Experimental Environment

4.1 Training Parameters

Critical parameters are used in the study to optimize the adaptive recommendation algorithm. The level of fuzziness in cluster assignments is determined by the fuzziness parameter (m), an essential component of the fuzzy k-means method. To accurately capture the relationships between users and financial goods, the NCF model is set up with three layers. The NCF uses an embedding size of 64 to balance computational economy and model performance. Users will obtain the most relevant financial alternatives according to their interests and investment habits because the system is designed to deliver the top five recommendations. The training-test split and the ability to provide top-N recommendations show that the system can scale and adapt to a growing dataset without losing accuracy or relevance.

4.2 Performance Assessment

For comparison, the existing algorithms such as ANFIS [19], PRS-MPT [20], IFCM [22], and K-L-KM [24] are considered for performance comparison. The metrics used for analysis are clustering quality evaluation using silhouette coefficient Davis Bouldin Index, followed by recommendation accuracy using mean square error, precision@k, recall@k, and processing time analysis.

4.2.1 Clustering Quality

One of the most important metrics for assessing clustering quality, the Silhouette coefficient and Davis Bouldin Index (*DBI*), is utilized to analyze financial data for adaptive recommendations. This will be useful in evaluating the clusters' quality, essential for grouping financial users based on their preferences.

A. Silhouette coefficient

The silhouette coefficient is an evaluation tool for user interest pattern grouping, and it is helpful in more extensive data on intelligent investment strategies. This coefficient measures cluster coherence, and a higher value suggests that the user behavior clusters are more distinct and well-defined. This statistic verifies that users are appropriately grouped according to their financial routines and preferences; it can take values between -1 and 1, with close to +1 signifying a high separation across clusters.

$$s(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))} \quad (21)$$

where $a(i)$ in Eq. (21) is the average distance between the point i and all other points in the same cluster, and $b(i)$ is the minimum average distance from point i to points in a different cluster. Improved and adaptive financial recommendation is the final result of the research's usage of the Silhouette coefficient to verify that fuzzy K-means clustering are good at capturing complex user patterns.

The dotted lines show in Fig. 4 shows that the average silhouette score plotted with cluster label and impact values. This evaluation provides significant and well-separated clusters, resulting in more accurate and adaptable financial recommendations. The silhouette score validates clustering methods, which are essential to adapting financial advice to particular user profiles in this research. Cluster 2 has the highest average silhouette score of 0.690, making it the most coherent and well-separated adaptive financial recommendation system user group. This suggests that users in this cluster share identical financial interest patterns, improving

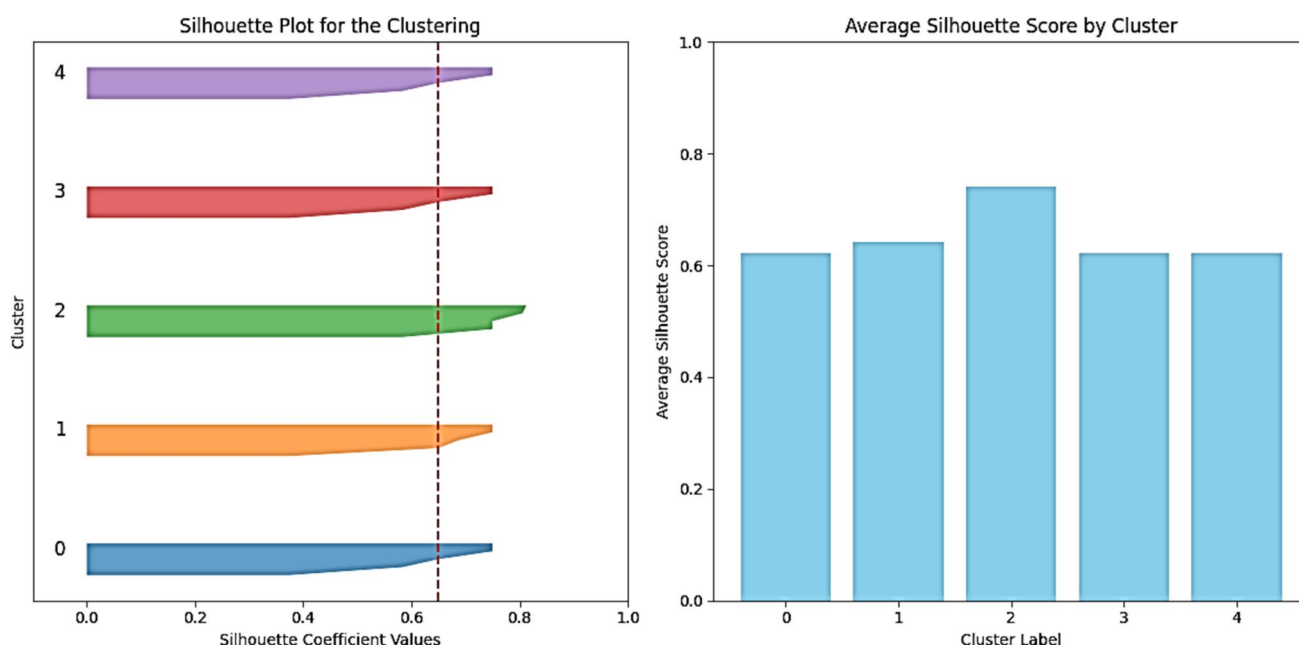


Fig. 4 Clustering quality using silhouette coefficient

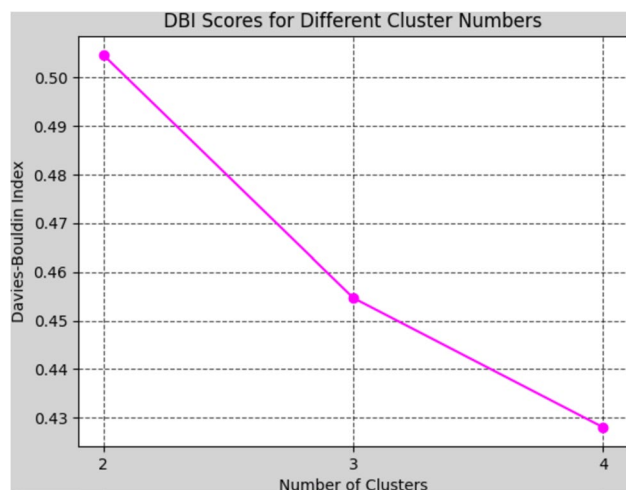


Fig. 5 Clustering quality using Davies–Bouldin index scores for different clusters

recommendation accuracy and relevance. An elevated average silhouette score throughout clusters indicates well-defined and distinctive user profiles, allowing the computer to make more personalized and successful financial suggestions based on data patterns.

B. Davies–Bouldin Index

The Davies–Bouldin Index measures cluster separation and compactness, making it valuable for clustering

performance evaluation. Low *DBI* result shows well-formed clusters and distinct user segments, suggesting that the FNFinRec system can target them. High *DBI* indicate that user segments are not diverse enough, requiring clustering redesign to analyze the user interest patterns based on the changing financial trends and services among users with more features.

The results of the Davies–Bouldin Index (DBI) shown in Fig. 5 demonstrated different FNFinRec cluster topologies shown in this picture. The DBI measures cluster separation and compactness. Lower values indicate that the clusters are better defined and more distinct. As the number of clusters increases, the graph probably shows that DBI scores decrease, which means that the clustering quality is improving. The ideal number of clusters to utilize for user-specific financial behavior and preference segmentation can be found with the help of this visualization. If the recommendation system is to adapt financial recommendations to different user segments successfully, then it is essential to analyze DBI scores.

4.2.2 Recommendation Accuracy

A. Mean square error

For each model $r_{u_r,i}$, provides the actual feedback and $\hat{r}_{u_r,i}$ indicates the predicted feedback for each user-financial service pair in Eq. (22) is given below:

$$MSE = \frac{1}{|R|} \sum_{(u_r,i) \in R} (r_{u_r,i} - \hat{r}_{u_r,i})^2 \quad (22)$$

Improved performance is shown by lower MSE values. To learn how model improvements result in better suggestions, it is helpful to monitor changes in MSE over iterations and across various models. When dealing with financial applications, this statistic comes in handy because bigger forecast errors might result in significant losses. It is necessary to refine the model if the MSE is large since it shows that the predictions made by the model $\hat{r}_{u_r,i}$ are considerably lower than the actual values. On the other hand, if the MSE is low, it means the model better predicts financial recommendations based on user interests. To ensure the proposed system gives accurate financial recommendations, it provides a quantitative way to assess and contrast how various models perform.

The suggested FNFinRec is one of several recommendation algorithms compared here with their MSE metric shown in Fig. 6. This proves that FNFinRec is more accurate than competing algorithms when it comes to making financial recommendations.

B. Precision@k

Compared to previous algorithms, FNFinRec shows better precision across a range of k values shown in Fig. 7, suggesting that a larger percentage of the top- k suggestions are relevant to users. As k increases, the system's accuracy may decrease, showing that there is trade-off between the quantity and quality of recommendations. This graph highlights the success of FNFinRec's neural collaborative filtering and fuzzy k-means clustering combination, which leads to more accurate and user-satisfying recommendations for financial products.

C. Recall@k

FNFinRec consistently obtains better recall, demonstrating that it can include a more substantial number of relevant financial goods in its top- k recommendations, regardless of the value of k . This suggests that more relevant items are caught as the suggestion list extends, since memory improves as k increases. The strength of FNFinRec is in its comprehensive financial advice, which ensures that users don't miss out on helpful products or services. Algorithms may differ in their recall growth rates, which could be a sign of how well they can find relevant items outside of the top few recommendations as given in Fig. 8. If users have frequently changing interest pattern, they are less likely to pass up useful financial services and products.

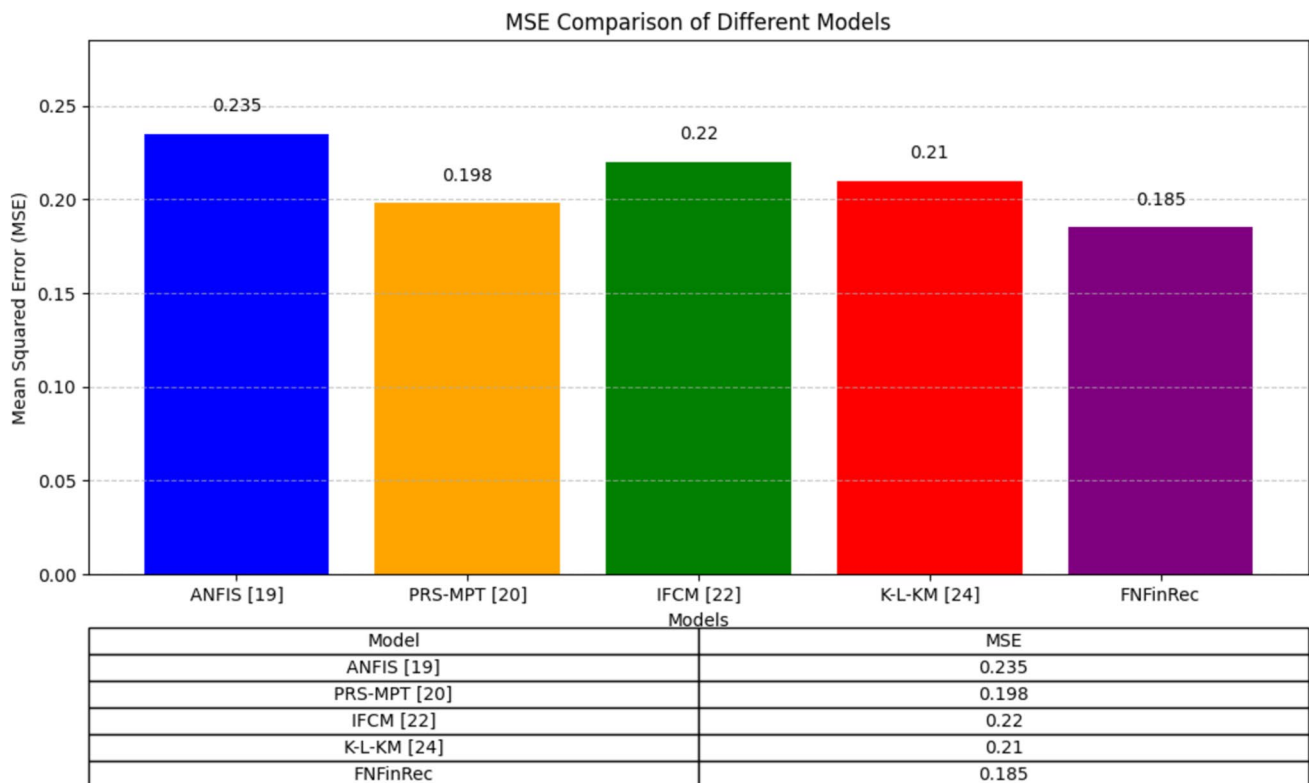


Fig. 6 Recommendation accuracy using mean average precision

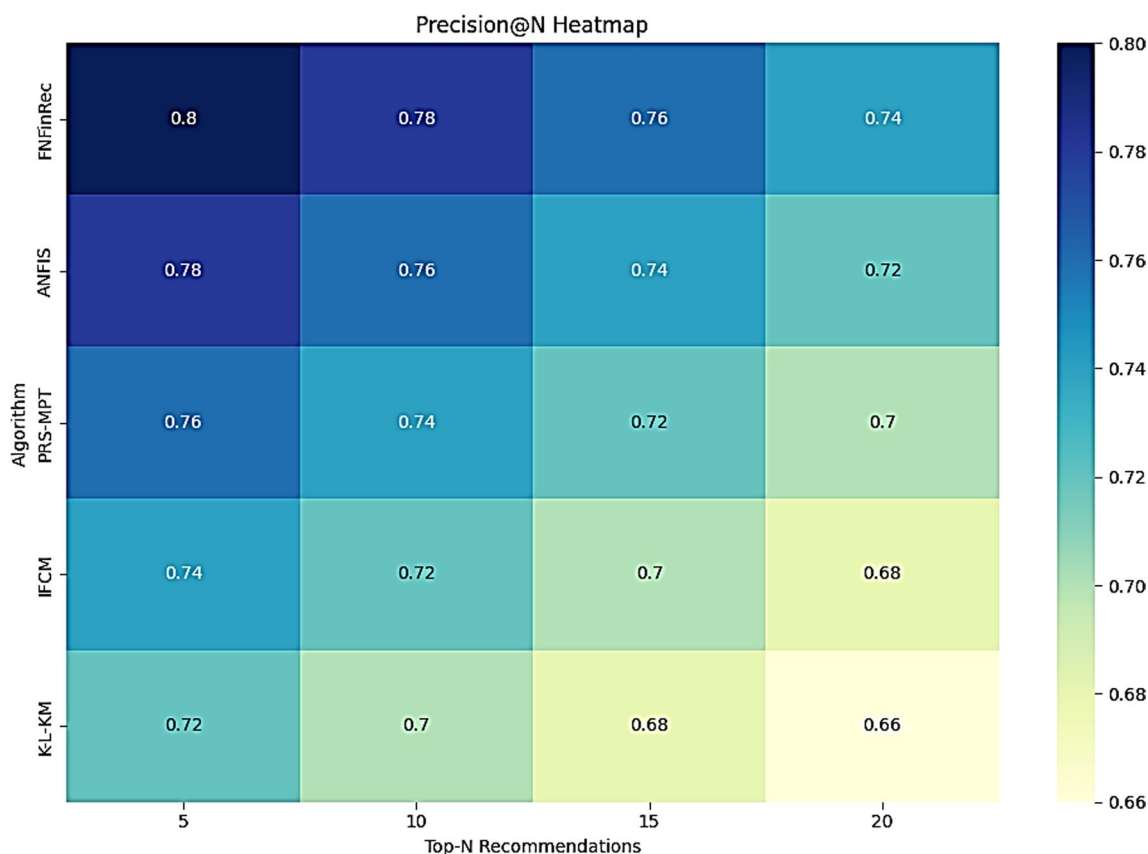


Fig. 7 Recommendation accuracy using Precision@k

4.2.3 Processing Time Analysis

The ability to update user interest patterns in real time and provide timely and appropriate recommendations is made possible by quicker processing speeds in financial data analysis. Because of its adaptability, the system can swiftly adjust to shifting user interests and market circumstances, resulting in better, more personalized investment recommendations. In financial data research, processing time analysis includes comparing the execution times of various algorithms and systems for preprocessing data, MapReduce operations, and clustering, among other tasks. The research found that processing times can vary greatly across FNFinRec, ANFIS, PRS-MPT, IFCM, and K-L-KM. If ANFIS requires 60 min to process data, while IFCM only takes 40 min, then IFCM must be more efficient as shown in Fig. 9. Decisions on the performance and scalability of the system can be guided by an analysis of these periods, which helps determine the algorithms that are most efficient for managing massive amounts of financial data. In earlier similarity calculation methods, the user similarity calculated is $O(|u|^2)$ and from this proposed approach mitigates the complexity to linear form.

5 Conclusion

The adaptive financial recommendation system processes user data in real-time to provide individualized suggestions, driven by Fuzzy K-means clustering and NCF. The rapid progression of user preferences and marketplace dynamics makes this an especially pertinent consideration in the world of finance. An adaptive financial recommendation system that offers real-time individualized investment advice utilizing Fuzzy K-means clustering and neural collaborative filtering (NCF) is presented in the research work. In the dynamic financial sector, where user interests and financial conditions are always shifting, this proposed FNFinRec algorithm excels. The solution outperforms conventional approaches by incorporating sophisticated algorithms, which improve the precision and customization of financial suggestions. Dealing with the immense amount and complexity of financial big data, keeping computing efficient, and adjusting with evolving s are all major challenges resolved using this algorithm. Incorporating more sophisticated deep learning methods, improving real-time flexibility, and fixing the issue for

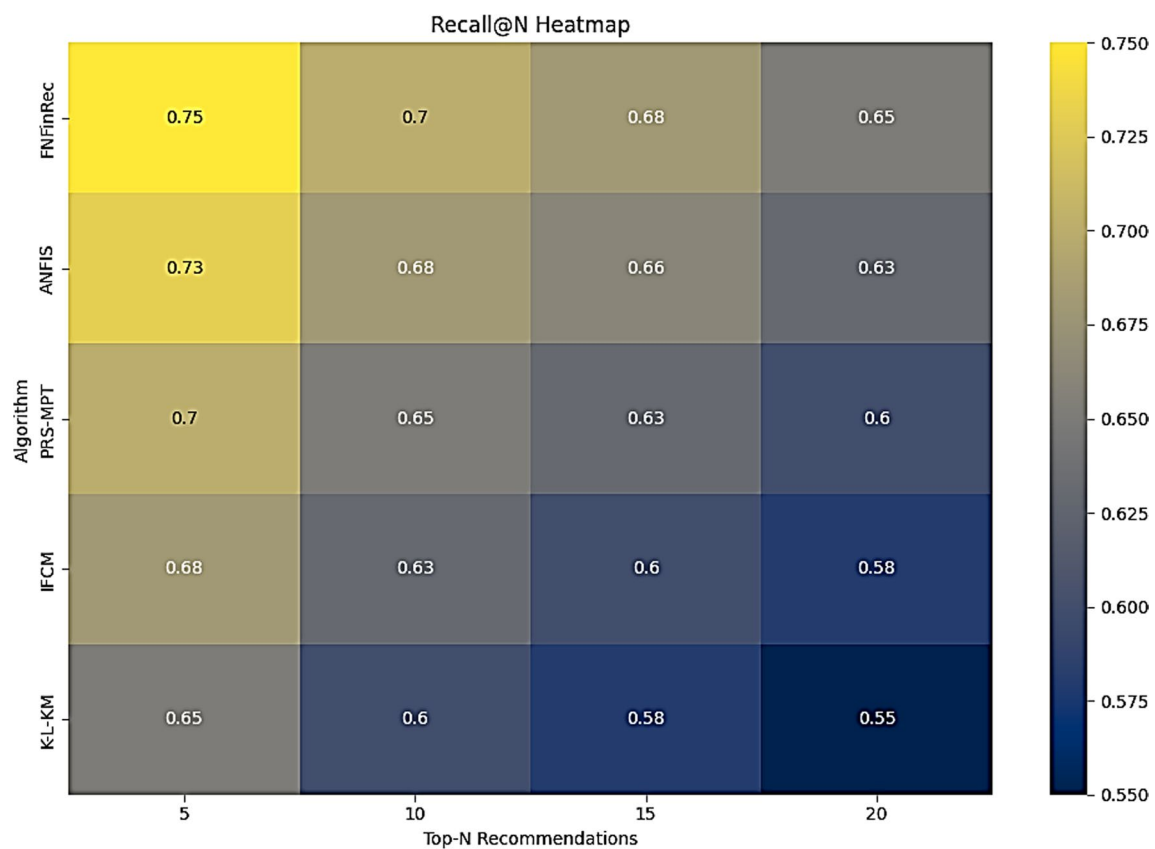
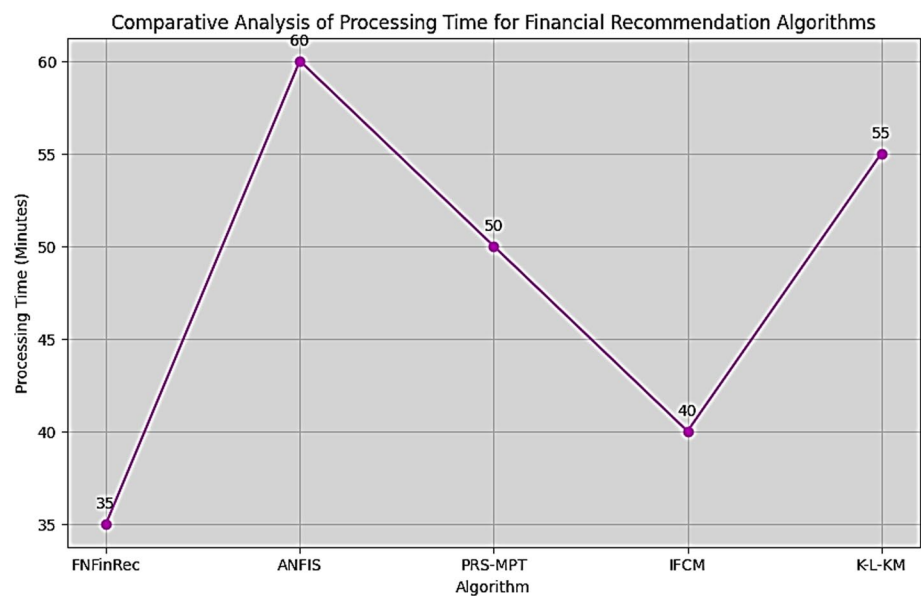


Fig. 8 Recommendation accuracy using Recall@k

Fig. 9 Processing time analysis



adaptive behavior for new users are all potential areas are focused with the user and financial service matrix and clustering quality analysis. Regardless of these issues, the system is a huge step forward in financial innovation and

shows promise in providing more personalized and precise financial recommendation.

Further research might look into making AI systems that are easier to understand and interact with more external data

sources that are updated in real-time, such as social media and news. It might also broaden the system's coverage to include more financial services and goods.

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Declarations

Conflict of interest It is declared by the authors that this article is free of conflict of interest.

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